B6 Condensed Matter Physics, 2025 — Model Solutions

Question 1 — X-ray diffraction from CaF_2

Problem. CaF_2 is fcc with Ca at $(0,0,0)$ and F at $\frac{1}{4}(1,1,1)$, $\frac{3}{4}(1,1,1)$. Define $S(\mathbf{G})$, derive S_{hkl}, identify the first two strong powder lines at $2\theta = 28.27°, 47.00°$ ($\lambda = 0.154\,06\,\text{nm}$), and estimate I_{111}/I_{220}.

(a) Definition and use of the structure factor

Sum of basis scatterers, weighted by atomic form factor $f_j(\mathbf{G})$ (Fourier transform of electron density, $f_j(0) = Z_j$):

$$S(\mathbf{G}) \;=\; \sum_{j\in\text{basis}} f_j(\mathbf{G})\, e^{i\mathbf{G}\cdot\mathbf{r}_j}. \tag{1}$$

Kinematic (first Born) approximation: $I(\mathbf{G}) \propto |S(\mathbf{G})|^2$ (Lorentz/polarisation/multiplicity ignored). Reflections with $S = 0$ are systematically absent.

For cubic a, $\mathbf{G}_{hkl} = (2\pi/a)(h,k,l)$. Laue $\mathbf{k}' - \mathbf{k} = \mathbf{G}$ for elastic scattering is equivalent to $\lambda = 2d_{hkl}\sin\theta$ with $d_{hkl} = 2\pi/|\mathbf{G}_{hkl}|$.

(b) Structure factor of CaF_2

Fcc Bravais factor:

$$F_{\text{fcc}}(hkl) \;=\; 1 + e^{i\pi(h+k)} + e^{i\pi(k+l)} + e^{i\pi(h+l)}, \tag{2}$$

which is 4 for h, k, l all-even or all-odd, else 0. With basis at $(0,0,0)$ and $\frac{1}{4}(1,1,1), \frac{3}{4}(1,1,1)$, $\mathbf{G}\cdot\frac{1}{4}(1,1,1)a = \frac{\pi}{2}(h+k+l)$, so the basis sum is

$$f_{\text{Ca}} + f_{\text{F}}\Big[e^{i\pi(h+k+l)/2} + e^{i3\pi(h+k+l)/2}\Big] = f_{\text{Ca}} + 2f_{\text{F}}\cos\big[\tfrac{\pi}{2}(h+k+l)\big]\,.$$

Hence

$$\boxed{S_{hkl} \;=\; F_{\text{fcc}}(hkl)\Big[\,f_{\text{Ca}} + 2f_{\text{F}}\cos\!\big(\tfrac{\pi}{2}(h+k+l)\big)\Big]} \tag{3}$$

with $F_{\text{fcc}} = 4$ on allowed reflections.

Three regimes: $h+k+l = 4n$ gives $S \propto f_{\text{Ca}} + 2f_{\text{F}}$ (strong); $h+k+l = 4n+2$ gives $S \propto f_{\text{Ca}} - 2f_{\text{F}}$ (near-extinct since $f_{\text{Ca}} \approx 2f_{\text{F}}$); $h+k+l$ odd gives $S \propto f_{\text{Ca}}$ (F ions cancel).

(c) Indexing the powder pattern and lattice constant

002: $h+k+l = 2$, so $S_{002} \propto f_{\text{Ca}} - 2f_{\text{F}} \approx 0$ (near-extinct). 111: $S_{111} \propto f_{\text{Ca}}$ (large).

Bragg + cubic:

$$\sin^2\theta \;=\; \frac{\lambda^2}{4a^2}\,(h^2+k^2+l^2). \tag{4}$$

$\theta_1 = 14.135°$, $\theta_2 = 23.50°$: $\sin^2\theta_1 = 0.05963$, $\sin^2\theta_2 = 0.15903$, ratio $2.667 \approx 8/3$. Assign 111 ($h^2+k^2+l^2 = 3$) and 220 ($= 8$); 200 skipped (weak, same argument as 002).

$$a = \frac{\lambda\sqrt{3}}{2\sin\theta_{111}} = \frac{0.15406 \times 1.7321}{2 \times 0.24417}\,\text{nm} \approx 0.5466\,\text{nm}.$$

Cross-check from 220: $a \approx 0.5464\,\text{nm}$.

$$\boxed{\text{First two strong reflections: 111 and 220,} \quad a \approx 0.547\,\text{nm.}} \tag{5}$$

(d) Form-factor graphs and intensity ratio of 111 versus 220

$f(G)$ is the Fourier transform of the ionic electron density: starts at $f(0) = Z_{\text{ion}}$ (here $f_{\text{Ca}^{2+}}(0) = 18$, $f_{\text{F}^-}(0) = 10$) and decays monotonically (closed shells).

At $a = 5.466\,\text{Å}$:

$$G_{111} = \frac{2\pi\sqrt{3}}{a} \approx 1.99\,\text{Å}^{-1}, \qquad G_{220} = \frac{2\pi\sqrt{8}}{a} \approx 3.25\,\text{Å}^{-1}. \tag{6}$$

Reading off: $f_{\text{Ca}}(G_{111}) \approx 14$, $f_{\text{F}}(G_{111}) \approx 7$; $f_{\text{Ca}}(G_{220}) \approx 11.5$, $f_{\text{F}}(G_{220}) \approx 5.5$. Drop the common $F_{\text{fcc}} = 4$:

$$S_{111} = f_{\text{Ca}} + 2f_{\text{F}}\cos(3\pi/2) = f_{\text{Ca}} \approx 14, \qquad S_{220} = f_{\text{Ca}} + 2f_{\text{F}} \approx 22.5.$$

$$\frac{I_{111}}{I_{220}} = \frac{14^2}{22.5^2} \approx 0.39.$$

$$\boxed{I_{111}/I_{220} \approx 0.4.} \tag{7}$$

At 220 all three ions add coherently; at 111 the two F ions cancel and only Ca contributes.

Question 2 — 1D tight-binding model

Problem. Chain of N atoms, $\langle n|\hat{H}|n\rangle = \mathcal{E}_0$, $\langle n|\hat{H}|n \pm 1\rangle = -t$, PBC. (a) Find $E(k)$ and $g(E)$. (b) Half-filled (monovalent) molar C and χ; comment on divalent. (c) Add $\langle n|\hat{H}|n \pm 2\rangle = -t/2$; modified $E(k)$, $g(E)$.

(a) Band $E(k)$, bandwidth, and density of states

Bloch ansatz $|k\rangle = N^{-1/2}\sum_n e^{ikna}|n\rangle$, PBC fixes $k = 2\pi m/(Na)$, N values in $(-\pi/a, \pi/a]$. Acting with $\hat{H}$:

$$\hat{H}|k\rangle = \frac{1}{\sqrt{N}}\sum_n (\mathcal{E}_0 - te^{-ika} - te^{ika})e^{ikna}|n\rangle = (\mathcal{E}_0 - 2t\cos ka)|k\rangle.$$

$$\boxed{E(k) = \mathcal{E}_0 - 2t\cos ka, \qquad A = \mathcal{E}_0, \quad B = -2t.} \tag{8}$$

$$\boxed{W = 4t.} \tag{9}$$

DOS (factor 2 spin, factor 2 from $\pm k$, k-density $Na/2\pi$):

$$g(E) = \frac{2Na}{\pi}\left|\frac{\mathrm{d}k}{\mathrm{d}E}\right| = \frac{2Na}{\pi} \cdot \frac{1}{|2ta\sin ka|}.$$

Invert dispersion:

$$\cos ka = \frac{\mathcal{E}_0 - E}{2t}.$$

Apply $\sin^2 = 1 - \cos^2$:

$$\sin^2 ka \;=\; 1 - \left(\frac{E-\mathcal{E}_0}{2t}\right)^2.$$

Take positive root:

$$|\sin ka| \;=\; \sqrt{1-\big[(E-\mathcal{E}_0)/(2t)\big]^2}.$$

Substitute into denominator:

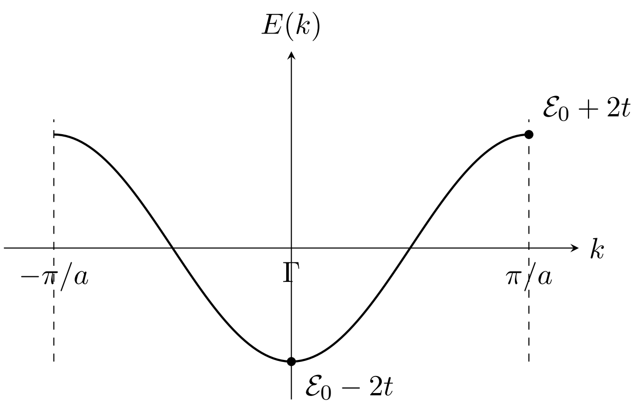
$$|2ta\sin ka| \;=\; 2ta\sqrt{1-\big[(E-\mathcal{E}_0)/(2t)\big]^2}.$$

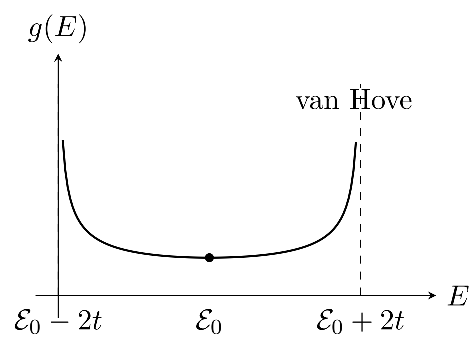
Combine with prefactor $2Na/\pi$:

$$g(E) \;=\; \frac{N}{\pi t}\,\frac{1}{\sqrt{1-\big[(E-\mathcal{E}_0)/(2t)\big]^2}} \;=\; \frac{\alpha N}{\sqrt{1-\beta(E-\mathcal{E}_0)^2}}, \tag{10}$$

$$\boxed{\alpha = \frac{1}{\pi t}, \qquad \beta = \frac{1}{4t^2}.} \tag{11}$$

Integrable $1/\sqrt{}$ van Hove divergences at $E = \mathcal{E}_0 \pm 2t$.





(b) Heat capacity and susceptibility for a half-filled band

Monovalent: N electrons in $2N$ states $\Rightarrow$ half-filled, $E_F = \mathcal{E}_0$. From (10), $g(E_F) = N/(\pi t)$.

Sommerfeld:

$$C_V \;=\; \frac{\pi^2}{3}\,g(E_F)\,k_B^2\,T.$$

Substitute $g(E_F) = N/(\pi t)$:

$$C_V \;=\; \frac{\pi^2}{3}\,\frac{N}{\pi t}\,k_B^2\,T.$$

Cancel π:

$$C_V \;=\; \frac{\pi}{3}\,Nk_B\,\frac{k_BT}{t}.$$

Per mole ($N = N_A$):

$$C \;=\; \frac{\pi}{3}\,R\,\frac{k_BT}{t}, \qquad \boxed{\xi = 1, \qquad \eta = \pi/3.} \tag{12}$$

Pauli susceptibility:

$$\chi_P = \mu_0 \mu_B^2 g(E_F)/V.$$

Per mole:

$$\boxed{\chi \;=\; \mu_0 \mu_B^2 \,\frac{N_A}{\pi t} \;=\; \frac{\mu_0 \mu_B^2 R}{\pi t k_B}.} \tag{13}$$

Temperature-independent.

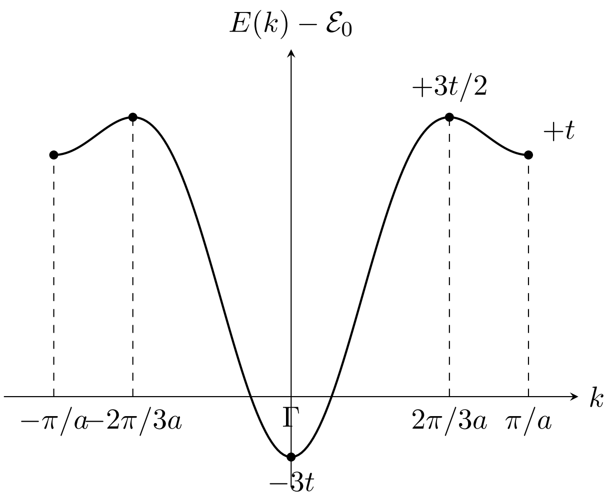
Divalent. $2N$ electrons fill the band; μ in gap above. Insulator: $C_{\rm el} \propto e^{-E_g/k_BT}$ (phonon T^3 dominates), $\chi_P \to 0$ (only small Landau/core diamagnetism).

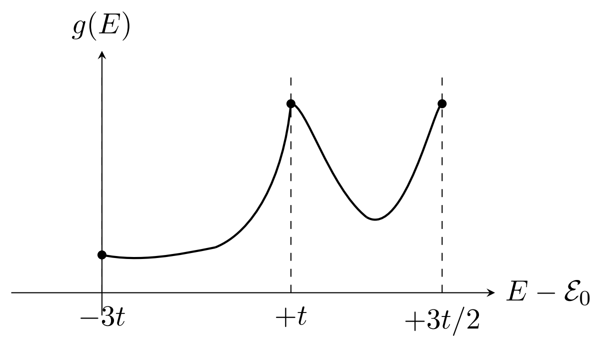
(c) Adding second-neighbour hopping

Second-neighbour adds $-(t/2)(e^{-2ika} + e^{2ika}) = -t\cos 2ka$:

$$\boxed{E(k) \;=\; \mathcal{E}_0 - 2t\cos ka - t\cos(2ka).} \tag{14}$$

Extrema: $\mathrm{d}E/\mathrm{d}k = 2ta\sin ka(1 + 2\cos ka) = 0 \Rightarrow ka = 0, \pm 2\pi/3, \pm\pi/a$. Values: $E(0) = \mathcal{E}_0 - 3t$ (global min), $E(\pm 2\pi/3a) = \mathcal{E}_0 + 3t/2$ (global max), $E(\pm\pi/a) = \mathcal{E}_0 + t$ (local min).





Three van Hove singularities, at $E = \mathcal{E}_0 - 3t, +t, +3t/2$.

Question 3 — Semiconductors and the Hall effect

Problem. Define intrinsic/extrinsic semiconductors; derive $n(T) \propto T^{\alpha} e^{-\eta E_g/k_BT}$ and give $p(T)$. Define and measure R_H; use $R_H(300)/R_H(350) = 28$ in undoped Si to get E_g. Estimate donor Bohr radius and ionisation energy in P-doped Si ($\epsilon_r = 11.7$, $m^* = 1.09\, m_e$); sketch $\ln R_H$ vs $1/T$.

(a) Intrinsic and extrinsic semiconductors

Intrinsic: pure host, valence full / conduction empty at $T = 0$, gap E_g. Thermal pair excitation gives $n = p$. *Extrinsic*: dopants put levels in the gap. Donors (e.g. P in Si) sit $E_d \ll E_g$ below E_c and donate electrons; acceptors sit above E_v and accept (creating holes). $n \neq p$ in general; majority carriers come from impurities.

(b) Conduction-band electron density $n(T)$

Parabolic band:

$$g_c(E) \;=\; \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}\sqrt{E-E_c}. \tag{15}$$

Non-degenerate (μ in gap): use Maxwell–Boltzmann tail. With $x=(E-E_c)/k_BT$ and $\int_0^\infty \sqrt{x}e^{-x}\mathrm{d}x = \sqrt{\pi}/2$,

$$n(T) \;=\; N_c(T)\,e^{-(E_c-\mu)/k_BT}, \qquad N_c(T) \;=\; 2\left(\frac{m_e^*k_BT}{2\pi\hbar^2}\right)^{3/2}. \tag{16}$$

Analogously for holes:

$$p(T) \;=\; N_v(T)\,e^{-(\mu-E_v)/k_BT}, \qquad N_v(T) = 2\left(\frac{m_h^*k_BT}{2\pi\hbar^2}\right)^{3/2}. \tag{17}$$

Mass action: $np = N_cN_ve^{-E_g/k_BT} \propto T^3e^{-E_g/k_BT}$. Intrinsic neutrality $n=p$:

$$n(T) = p(T) = \sqrt{N_cN_v}\,e^{-E_g/(2k_BT)} \propto T^{3/2}e^{-E_g/(2k_BT)}.$$

$$\boxed{\alpha = 3/2, \qquad \eta = 1/2.} \tag{18}$$

$$\boxed{p(T) = n(T) \propto T^{3/2}\,e^{-E_g/(2k_BT)}.} \tag{19}$$

The $E_g/2$ in the Boltzmann factor reflects μ pinned mid-gap.

(c) Hall coefficient: definition, measurement, and E_g from undoped Si

Definition. With j_x, B_z, transverse field E_y balances the Lorentz force in steady state:

$$R_H \;=\; \frac{E_y}{j_xB_z}. \tag{20}$$

Single carrier of charge q, density n:

$$\boxed{R_H \;=\; \frac{1}{nq}.} \tag{21}$$

Sign distinguishes electrons ($R_H<0$) from holes ($R_H>0$).

Measurement. Thin rectangular bar; force longitudinal current I, measure transverse Hall voltage V_H with high-impedance voltmeter. Width w, thickness d: $R_H = V_Hd/(IB)$. Reverse B and average to remove offsets.

Why electrons dominate in Si. $\mu_e \approx 1400\,\mathrm{cm^2V^{-1}s^{-1}} \gg \mu_h \approx 450\,\mathrm{cm^2V^{-1}s^{-1}}$; in two-carrier intrinsic Si, $R_H = (p\mu_h^2 - n\mu_e^2)/[e(n\mu_e+p\mu_h)^2] < 0$, electron-dominated.

Extracting E_g. Use $R_H \propto 1/n$ with $n \propto T^{3/2}e^{-E_g/(2k_BT)}$:

$$\frac{R_H(300)}{R_H(350)} = \left(\frac{350}{300}\right)^{3/2}\exp\!\left[\frac{E_g}{2k_B}\left(\frac{1}{300}-\frac{1}{350}\right)\right] = 28. \tag{22}$$

Evaluate the prefactor:

$$\left(\frac{350}{300}\right)^{3/2} \;\approx\; 1.260.$$

Isolate the exponential:

$$\exp\!\left[\frac{E_g}{2k_B}\left(\frac{1}{300}-\frac{1}{350}\right)\right] \;=\; \frac{28}{1.260} \;\approx\; 22.2.$$

Take the logarithm:

$$\frac{E_g}{2k_B}\left(\frac{1}{300} - \frac{1}{350}\right) \;=\; \ln 22.2 \;\approx\; 3.10.$$

Evaluate the temperature factor:

$$\frac{1}{300} - \frac{1}{350} \;=\; 4.762 \times 10^{-4}\,\mathrm{K}^{-1}.$$

Solve for E_g:

$$E_g \;=\; \frac{2k_B \times 3.10}{4.762 \times 10^{-4}}\,\mathrm{J} \;\approx\; 1.80 \times 10^{-19}\,\mathrm{J}.$$

$$\boxed{E_g \;\approx\; 1.12\,\mathrm{eV}.} \tag{23}$$

(d) Donor Bohr radius and ionisation energy in P-doped Si

Hydrogenic in dielectric medium with effective mass: $\epsilon_0 \to \epsilon_0\epsilon_r$, $m_e \to m_e^*$:

$$a^* = \epsilon_r \frac{m_e}{m_e^*} a_0, \qquad E_R^* = \frac{1}{\epsilon_r^2}\frac{m_e^*}{m_e} E_R. \tag{24}$$

With $\epsilon_r = 11.7$, $m_e^*/m_e = 1.09$, $a_0 = 0.529\,\text{Å}$, $E_R = 13.6\,\mathrm{eV}$, evaluate the Bohr radius:

$$a^* \;=\; \frac{11.7}{1.09} \times 0.529\,\text{Å} \;\approx\; 5.68\,\text{Å}.$$

Evaluate the Rydberg:

$$E_R^* \;=\; \frac{1.09 \times 13.6}{11.7^2}\,\mathrm{eV} \;\approx\; 0.108\,\mathrm{eV}.$$

$$\boxed{a^* \approx 0.57\,\mathrm{nm}, \qquad E_d \approx 0.11\,\mathrm{eV}.} \tag{25}$$

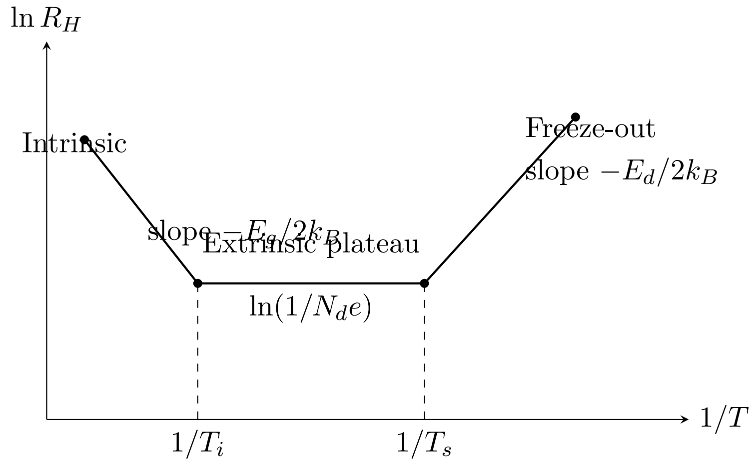
$a^* \gg a_{\mathrm{Si}}$ justifies the dielectric continuum picture; experimental $E_d \approx 45\,\mathrm{meV}$ for P:Si, the discrepancy from anisotropic mass and central-cell corrections.

(e) Sketch of $\ln R_H$ versus $1/T$ for the P-doped sample

Three regimes (rising T):

1. Freeze-out ($k_BT \ll E_d$): $n \propto e^{-E_d/(2k_BT)}$, $\ln R_H$ rises with $1/T$, slope $\sim -E_d/2k_B$.
2. Saturation: all donors ionised, $n \approx N_d$, $\ln R_H$ flat.
3. Intrinsic: $n_i > N_d$, $\ln R_H \propto E_g/(2k_BT)$, steep slope $-E_g/2k_B$.

Crossover when $\sqrt{N_cN_v}e^{-E_g/(2k_BT)} = N_d$; for $N_d = 6.25 \times 10^{18}\,\mathrm{m}^{-3}$, $E_g = 1.12\,\mathrm{eV}$, $T \sim 500\,\mathrm{K}$.



Question 4 — Heat capacity, Debye and magnetic contributions

Problem. Compare Einstein/Debye assumptions; derive low-T $C_{\text{Debye}} = xpR(T/T_D)^y$. For N non-interacting $S = 1/2$ spins derive $Z = [2\cosh z]^N$, hence χ and C_{mag}. Compare $C_{\text{mag}}/C_{\text{Debye}}$ at 10 K, $B = 0.1$ T and 10 T, with $T_D = 200$ K, $p = 10$, one magnetic atom per formula unit.

(a) Einstein versus Debye models

Both: harmonic phonons, independent modes, Bose–Einstein occupation, Dulong–Petit limit $3Nk_B$ at high T.

Differ: Einstein gives all $3N$ modes $\omega = \omega_E$ (good for optical, zone-centre); $C \propto e^{-\hbar\omega_E/k_BT}$ at low T — too rapid. Debye uses linear acoustic dispersion $\omega = c_s k$ up to cutoff ω_D (mode count = $3N$); captures long-wavelength modes correctly, gives $C \propto T^3$.

(b) Debye internal energy and low-T heat capacity

Debye DOS (3 acoustic branches, N_a atoms):

$$g(\omega) \;=\; \frac{9N_a}{\omega_D^3}\,\omega^2, \qquad 0 \le \omega \le \omega_D. \tag{26}$$

$$U(T) = \int_0^{\omega_D} \mathrm{d}\omega\, g(\omega)\frac{\hbar\omega}{e^{\hbar\omega/k_BT} - 1}.$$

Substitute $x = \hbar\omega/k_BT$, $T_D = \hbar\omega_D/k_B$:

$$U(T) \;=\; 9N_ak_BT\left(\frac{T}{T_D}\right)^3\int_0^{T_D/T}\frac{x^3\,\mathrm{d}x}{e^x - 1}.$$

Low T, extend the upper limit to ∞:

$$U(T) \;\approx\; 9N_ak_BT\left(\frac{T}{T_D}\right)^3\int_0^{\infty}\frac{x^3\,\mathrm{d}x}{e^x - 1}.$$

Use $\int_0^\infty x^3/(e^x - 1)\,\mathrm{d}x = \pi^4/15$:

$$U(T) \;\approx\; 9N_ak_BT\left(\frac{T}{T_D}\right)^3 \cdot \frac{\pi^4}{15}.$$

Simplify:

$$U(T) \approx \frac{3\pi^4}{5} N_a k_B T \left(\frac{T}{T_D}\right)^3.$$

Differentiate with respect to T:

$$C_V = \frac{12\pi^4}{5} N_a k_B \left(\frac{T}{T_D}\right)^3.$$

Per mole of formula units, $N_a = pN_A$:

$$\boxed{C_{\text{Debye}}(T) = \frac{12\pi^4}{5} pR \left(\frac{T}{T_D}\right)^3, \qquad x = \frac{12\pi^4}{5},\ y = 3,\ T_D = \frac{\hbar\omega_D}{k_B}.} \tag{27}$$

(c) Paramagnet: partition function, susceptibility, and C_{mag}

$S = 1/2$, $g_s = 2$: $\epsilon_\pm = \mp\mu_B B$, so

$$Z_1 = 2\cosh z, \qquad z \equiv \frac{\mu_B B}{k_B T}.$$

Independent spins:

$$Z = [2\cosh(z)]^N. \tag{28}$$

Free energy:

$$F = -Nk_B T \ln(2\cosh z).$$

Magnetisation:

$$M = -\frac{1}{V}\frac{\partial F}{\partial B} = \frac{N\mu_B}{V} \tanh z.$$

Curie limit $z \ll 1$, expand $\tanh z \approx z$:

$$M \approx \frac{N\mu_B}{V} z = \frac{N\mu_B^2 B}{V k_B T}.$$

Differentiate:

$$\chi(T) = \mu_0 \left.\frac{\partial M}{\partial B}\right|_{B=0} = \frac{\mu_0 N\mu_B^2}{V k_B T}. \tag{29}$$

Internal energy:

$$U_{\text{mag}} = -N\mu_B B \tanh z.$$

Use $\mathrm{d}z/\mathrm{d}T = -z/T$ and $\mathrm{d}(\tanh z)/\mathrm{d}z = \operatorname{sech}^2 z$:

$$\frac{\mathrm{d}U_{\text{mag}}}{\mathrm{d}T} = -N\mu_B B \operatorname{sech}^2 z \cdot \left(-\frac{z}{T}\right).$$

Simplify with $\mu_B B = zk_B T$:

$$C_{\text{mag}} = N\mu_B B \operatorname{sech}^2 z \cdot \frac{z}{T} = Nk_B z^2 \operatorname{sech}^2 z.$$

Per mole of spins:

$$\boxed{C_{\text{mag}}(T, B) = R\,[f(z)]^2, \qquad f(z) = \frac{z}{\cosh z}, \qquad z = \frac{\mu_B B}{k_B T}.} \tag{30}$$

Schottky form: $\propto z^2$ for $z \ll 1$, exponentially suppressed for $z \gg 1$, peak near $z \approx 1.2$.

(d) Numerical comparison and effect of strong interactions

At $T = 10\,\mathrm{K}$, evaluate the Debye prefactor:

$$C_{\mathrm{Debye}} \;=\; \frac{12\pi^4}{5}\cdot 10\cdot R\left(\frac{10}{200}\right)^3.$$

Numerically:

$$\frac{12\pi^4}{5}\cdot 10 \;\approx\; 233.78, \qquad \left(\frac{10}{200}\right)^3 \;=\; \frac{1}{8000}.$$

Combine:

$$C_{\mathrm{Debye}} \;\approx\; \frac{233.78}{8000}\,R \;\approx\; 0.02923\,R.$$

One mole of spins per mole of formula unit. With $\mu_B/k_B = 0.6717\,\mathrm{K\,T^{-1}}$:

(i) $B = 0.1\,\mathrm{T}$**:** evaluate z:

$$z \;=\; \frac{\mu_B B}{k_B T} \;=\; \frac{0.6717\times 0.1}{10} \;\approx\; 6.72\times 10^{-3}.$$

In the Curie limit $z \ll 1$:

$$C_{\mathrm{mag}} \;\approx\; Rz^2 \;\approx\; 4.51\times 10^{-5}\,R.$$

Ratio:

$$\boxed{\left.\frac{C_{\mathrm{mag}}}{C_{\mathrm{Debye}}}\right|_{0.1\,\mathrm{T}} \;\approx\; 1.5\times 10^{-3}.} \tag{31}$$

(ii) $B = 10\,\mathrm{T}$**:** evaluate z:

$$z \;=\; \frac{0.6717\times 10}{10} \;=\; 0.6717.$$

Evaluate $\cosh z$ and $f(z) = z/\cosh z$:

$$\cosh z \;\approx\; 1.232, \qquad f(z) \;\approx\; \frac{0.6717}{1.232} \;\approx\; 0.5453.$$

Hence:

$$C_{\mathrm{mag}} \;=\; R\,[f(z)]^2 \;\approx\; 0.2974\,R.$$

$$\boxed{\left.\frac{C_{\mathrm{mag}}}{C_{\mathrm{Debye}}}\right|_{10\,\mathrm{T}} \;\approx\; 10.} \tag{32}$$

At $10\,\mathrm{T}$ the Schottky term exceeds the phonon background by an order of magnitude — the basis of adiabatic-demagnetisation cooling.

Strongly interacting spins at $B = 0$. Exchange J generates an internal field; below $T_c \sim J/k_B$ the spins order. Expect:

- Lambda peak in $C_{\mathrm{mag}}(T, 0)$ at T_c (second-order transition).
- Magnon power law below T_c: $T^{3/2}$ (3D ferromagnet, $\omega \propto k^2$); T^3 (3D antiferromagnet, $\omega \propto k$).
- High-T tail $\propto 1/T^2$ from short-range correlations.
- Total entropy $\int (C_{\mathrm{mag}}/T)\mathrm{d}T = R\ln 2$ conserved (spin degeneracy).
- Curie–Weiss susceptibility $\chi \propto 1/(T - \Theta_W)$, $\Theta_W \gtrless 0$ for FM/AFM.